Apple Surface Pesticide Residue Detection Method Based on Hyperspectral Imaging

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Abstract. In order to study the rapid and effective non-destructive detection method of pesticide residues on apple surface, this paper uses hyperspectral imaging technology to verify the feasibility of pesticide residue detection on apple surface. 225 apple samples from two groups were collected to construct the discriminant models of two pesticide residues, i.e., chlorpyrifos and carbendazim. The Hough circle transformation technique was used to determine the Region of Interest (ROI) automatically, and the averaged spectral value of the ROI is calculated as the representative spectrum of the sample. Then the Savitzky-Golay smoothing method was used for spectral denoising. Finally, the discriminant modeling is performed on the whole band with five methods: linear discriminant analysis, linear support vector machine, K nearest neighbor, decision tree and subspace discriminant ensemble. Furthermore, feature band selection was carried out by the successive projection algorithm and subspace discriminant ensemble method, then discriminant models were constructed on the feature band using linear discriminant analysis, linear support vector machine and K nearest neighbor. The experimental results show that the classification accuracy in both the whole band and the selected feature band for the detection of pesticide residues can be up to 95%. For the prediction of pesticide residue concentration, the subspace discriminant ensemble method based on the full band performs better, in which chlorpyrifos pesticide concentration prediction accuracy of up to 95%. The results confirmed the feasibility and effectiveness of hyperspectral imaging to detect pesticide residues on apple surface.

Keywords: Non-destructive detection · Pesticide residues
Hyperspectral imaging · Subspace discriminant ensemble

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1 Introduction

Application of pesticides is increasing rapidly all over the world. Various pesticides are applied worldwide for crop protection to increase their quality, increase the yield as well as extend the storage time [1]. As people's living standards improve, consumers raise awareness of food safety issues. China is the country that uses the largest amount of pesticides in the world. Pesticides effectively reduce the hazards of pests, weeds and other diseases and improve the quality and yield of agricultural products [2]. Pesticide residues are unavoidable in the process of using pesticides. Pesticides directly attached to the surface of agricultural products can cause serious damage to human health. Control of pesticide residues has become an important part of food safety issues, and related issues of pesticide detection is also more and more important.

The application of hyperspectral imaging and computer vision technology in food processing and testing has become a hot research topic in the field of agriculture [3]. It has received widespread attention in food safety and quality testing and is widely used in the nondestructive testing of agricultural products.

Hyperspectral imaging technology is widely used in the detection of agricultural products diseases and quality [4, 5] (such as surface defects, sugar content, moisture content, solidity, internal defects and surface contamination) In 2006, Nicolaï et al. [6] proposed Non-destructive measurement of bitter pit in apple fruit using NIR hyperspectral imaging. In 2008, Zhao [7] detect subtle bruises on fruits with hyperspectral imaging. In 2011, Wang et al. [8] use hyperspectral imaging to detect external insect infestations in jujube fruit. In 2012, Rajkumar et al. [9] use hyperspectral imaging to study the banana fruit quality and maturity stages. Nanyam et al. [9] propose a decision-fusion strategy for fruit quality inspection using hyperspectral imaging. In 2013, Lorente et al. [11] use hyperspectral imaging to compare the ROC feature selection method for the detection of decay in citrus fruit. Haff et al. [12] detect fruit fly infestation with hyperspectral images of mangoes.

In contrast, Hyperspectral imaging technology is still in the experimental stage in pesticide residue detection. In 2008, Xue et al. [13] use Hyperspectral Imaging to detect pesticide residue on Navel Orange Surface. In 2010, Nansen et al. [14] use spatial structure analysis of hyperspectral imaging data to determine bioactivity of surface pesticide. Dai et al. [15] propose a nondestructive detection method for pesticide residue on Longan surface. In 2014, Liu et al. [16] use hyperspectral imaging to detect pesticide residues on Lingwu long jujubes' surface. In 2015, Wu et al. [17] use hyperspectral remote sensing to detect pesticide residues. Chen et al. [18] propose a novel technique of an unsupervised subpixel detection algorithm based on hyperspectral imaging for pesticide residue detection. In 2016, Nansen et al. [19] use hyperspectral imaging to characterize consistency of coffee brands. In 2017, Sun et al. [20] propose a quantitative detection method for lettuce leaves based on hyperspectral technique. Mohite et al. [21] apply hyperspectral techniques to the detection of grape pesticide (cyanomethacin) residues

At present, there is no report on the method of using hyperspectral technology to detect pesticide residues in apples. At the same time, hyperspectral image technology is still in the experimental stage for pesticide residue detection. Further research is needed to

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improve model accuracy, reduce data redundancy, select feature bands, etc. In this paper, we conducted an experimental study on the detection of apple pesticide residues based on near-infrared hyperspectral, using a variety of pesticide residue discriminant models to compare experimental results, and applied ensemble learning in the field of machine learning to experiments. The results verify that the method of using the hyperspectral imaging technology to detect pesticide residues on apple surfaces is feasible

2 Methods

2.1 Feature Band Selection Algorithm

2.1.1 Successive Projections Algorithm

The Successive Projections Algorithm (SPA) [22] is a novel variable selection strategy for multivariate calibration which uses simple operations in a vector space to minimize variable collinearity. SPA method starts from one feature of the data, then select a new feature at each iteration, until the required number of features is reached.

Let $\boldsymbol{X_{k(0)}}$ be the initial iteration vector and $\boldsymbol{N}$ be the number of variables to be extracted. The spectral matrix has a total of $\boldsymbol{J}$ columns.

Step 1: Before iteration begins, randomly select one column $\boldsymbol{j}$ of the spectral matrix and assign the $\boldsymbol{j}$th column of the data set to $\boldsymbol{x_j}$, denoted $\boldsymbol{X_{k(0)}}$

Step 2: The unselected column vector set denoted as S,

$$S = \{x_j, 1 \leq j \leq J, j \notin \{k(0), k(1), \cdots, k(n-1)\}\} \tag{1}$$

Step 3: Calculate the projection of $\boldsymbol{x_j}$ on the remaining column vectors respectively

$$Px_j = x_j - (x_j^T x_{k(n-1)}) x_{k(n-1)} (x_{k(n-1)}^T x_{k(n-1)})^{-1}, j \in S \tag{2}$$

Step 4: Calculate k(n)

$$k(n) = \arg(\max(||Px_j||), j \in S \tag{3}$$

Step 5: Calculate $\boldsymbol{x_j}$

$$x_j = Px_j, j \in S \tag{4}$$

Step 6: Let n increase by one, if n < N go back to Step 1 for next iteration

Step 7: The final extracted variables are $\boldsymbol{F}$:

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$$F = \{k(n); n = 0, \cdots N-1\} \tag{5}$$

You can also calculate the root mean square error (RMSE) by Multiple linear regression (MLS) method to find the optimal initial vector $\boldsymbol{X_{k(0)}}$ and number of features $\boldsymbol{N}$ by setting a range to $\boldsymbol{N}$.

2.1.2 Subspace Discriminant Ensemble

Ensemble learning is a machine learning method that accomplishes learning tasks by combining multiple learners [23]. Usually ensemble learning is used to achieve better results than a single learner, and ensemble learning is especially effective for "weak learners".

In the Subspace Discriminant Ensemble (SDE) method, the selected weak learner is discriminant learner, and combined with the random subspace method [24–30] to extract features to classify the samples. This method is more suitable for the processing of high dimensional data.

The algorithm details are as follows:

Input:

Training data $D = \{(X_1, y_1), (X_2, y_2), \ldots, (X_m, y_m)\}$
Base learning algorithm: δ
The number of weak learners: N
Subspace dimension: d

Algorithm steps:

For $t = 1, 2, ..., T$ do

$t = 1, 2, ..., T$

$F_t = RS(D, d)$

$D_t = Map_{F_t}(D)$

$h_t = \delta(D_t)$

End For

Output:

$$H(x) = \arg\max \sum_{t=1}^{T} S(h_t(Map_{Ft}(x)) = y), y \in Y$$

In the input of the algorithm, D is the training data, X are the features of the sample, y is the label of the sample.

In the steps of the algorithm.

$F_t = RS(D, d)$ can randomly pick d-dimensional features from D.
$D_t = Map_{F_t}(D)$ can map selected subspace features from D.
$h_t = \delta(D_t)$ use training data to train weak learners.

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In the output of the algorithm, $H(x)$ is the final classification result given by the Subspace Discriminant Ensemble method, where $S(x)$ is indicator function. If x is true, $S(x) = 1$, else $S(x) = 0$.

Considering that SDE selects the band for discriminative modeling every time, we can also apply this method to the selection of feature bands. Thus, by recording which bands are used by a well-performing classifier, these bands can be selected as feature bands.

2.2 Discriminant Modeling Algorithm

2.2.1 Linear Discriminant Analysis

The idea of linear discriminant analysis (LDA) is relatively simple. This method projects the sample to an optimal discriminant space. This discriminant space makes the projections of similar samples as close as possible, and the sample projections between classes and classes are as far away as possible, that is, there is a minimum intraclass distance and a maximum class spacing after projection to obtain the best separation.

2.2.2 Support Vector Machine

Support Vector Machine (SVM) classifies data by finding the best hyperplane that separates data points of one class from data points of another. The best hyperplane is the hyperplane with the largest margin between the two classes. The margin is the maximum width of the plate parallel to the hyperplane without internal data points. Linear SVM refers to a support vector machine method in which the kernel function is a linear function, and the method performs simple linear separation between classes.

2.2.3 K-Nearest Neighbor

K-Nearest Neighbor (KNN) classification algorithm is one of the simplest methods in data mining classification technology. The so-called K nearest neighbor is the meaning of the k nearest neighbors. It is said that each sample can be represented by its closest k neighbors.

2.2.4 Decision Tree

A decision tree is a decision analysis method that determines the feasibility of a project by determining the probability that the expected value of the net present value is greater than or equal to zero by constructing a decision tree based on the probability of occurrence of various conditions. It is a graphical method of using probability analysis intuitively. The depth of the control tree in this experiment is 4.

2.2.5 Subspace Discriminant Ensemble

In this experiment, we adopt a random subspace method. When specific to the experimental hyperspectral data, we randomly select n features as their characteristic subspaces in all features of the sample. For example, there are a total of 256 bands of information in the acquired hyperspectral data sample. For subspace discrimination integration methods, the specified subspace dimension is 128 and the number of weak learners is 30. When training each weak classifier, this method randomly selects 128 features from 256 features for classifying modeling, and trained a total of 30 weak classifiers. For test data, this method combines the classification results of all classifiers to give the final classification result.

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3 Data Collection and Preprocessing

3.1 Apple Sample Collection

The test sample was collected from Yangling apple farming demonstration station. The apple tree was a biennial dwarfing rootstock densely-woven spindle-shaped Fuji. 36 trees were selected as the test object after a thorough spray cleaning. The samples were divided into four groups: control group, chlorpyrifos treatment group, carbendazim treatment group, chlorpyrifos and carbendazim combination group, with 9 trees in each group, of which 3 was a treatment concentration, divided into low, medium, High three doses, the control group sprayed with clean water. The dilution ratios of chlorpyrifos were 1: 500, 1: 1000 and 1: 2000, and the dilution ratios of carbendazim were 1: 600, 1: 1200 and 1: 2400 in turn. The dose of mixed group corresponded to the other two doses (1: 500 + 1: 600, 1: 1000 + 1: 1200, 1: 2000 + 1: 2400).

The experimental data is divided into two batches of acquisition, the first batch was taken 12 h after spraying, the second batch was collected 96 h after spraying. Two batches of different treatment methods to obtain the number of test samples shown in Table 1. And data processing flow chart shown in Fig. 1

Table 1. Details of different treatment samples

	Chlorpyrifos	Carbendazim	Mixed	Control
12 h	27	27	27	54
96 h	21	25	27	26

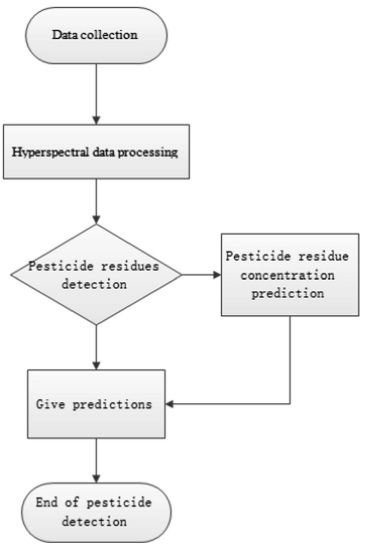


Fig. 1. The overall flow of data processing

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3.2 Hyperspectral Data Acquisition

After sample processing was completed, the hyperspectral data of the sample was acquired using ZOLIX HyperSIS-VNIR-PFH hyperspectral equipment. Data acquisition equipment shown in Fig. 2. The device is capable of capturing spectra in the 865.11 (nm) −1711.71 (nm) range, with a single hyperspectral image at 3.32 nm and a total of 256 imaging times in different bands. In order to obtain more complete information, the apple was on both sides of the imaging. Six apple samples were imaged each time.



Fig. 2. Hyperspectral imaging equipment

The imaging equipment corrects the reflectivity of hyperspectral images using white and black correction images, white calibration image using a polytetrafluoroethylene-based calibration plate (with reflectivity close to 1). While black correction was achieved by blocking the imaging lens (its reflectivity close to 0) [30]. Equation (6) is the reflectivity correction formula. Where $P_{(i,j)}$ is the value of the jth column of the ith row before the image is corrected and $P'_{(i,j)}$ is the corresponding value of the jth column of the ith row of the after the image is corrected. $D_{(i,j)}$ and $W_{(i,j)}$ represent the corresponding values of the black correction image and the white correction image in the ith column and the jth column, respectively.

$$P'_{ij} = \frac{P_{ij} - D_{ij}}{W_{ij} - D_{ij}} \tag{6}$$

The resulting imagery of the device includes hyperspectral source files (raw format) and their corresponding header files (hdr format). Hyperspectral source files are stored in Band Interleaved by Line format (BIL). Hyperspectral header files contain information about the source band's imaging band, storage format, and more. This experiment uses matlab 2015b software and Python language for experimental analysis.

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3.3 Hyperspectral Data Processing

After obtaining the hyperspectral image, in order to obtain the spectral information of the apple surface, a Region of Interest (ROI) is performed to obtain an image of the apple.

Detection of circles in images is an important task in computer vision and pattern recognition. Hough Transform (Hough Transform) is a method of boundary shape detection in image processing [31, 32]. It detects objects with specific shapes by a voting algorithm and is used to detect straight line segments, circles and ellipses. The basic idea of the Hough transform is to map the spatial shape information of the image into the parameter space. The constraint function of the image boundary point determines the value of the accumulator in the parameter space. The higher the value is, the bigger the probability that the boundary point forms the target shape. As a result, the problem of shape detection successfully transformed into the peak for the problem. Hough transform technique calculates boundary point information according to local measure, therefore, it has good fault tolerance and robustness to the circumstance that the boundaries are interrupted due to the noise of regional boundary or covered by other targets.

For circular detection, the three parameters of the circle, the coordinates of the circle center (x_0, y_0), and the length r_0 of the radius need to be determined in the image. For the circle in the image, the curve parameter can be written as (7). Where (x, y) is the point on the circumference.

$$(x - x_0)^2 + (y - y_0)^2 = r_0^2 \tag{7}$$

The first thing to do is to detect the edge of the image. Sobel operator can be used. Formula (8) is its matrix form, where Sx and Sy are the convolution templates in x direction and y direction respectively, K is the matrix of the target x and its neighborhood. The gradient of each point can be expressed as Gx in (9).

$$S_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, S_y = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}, K = \begin{pmatrix} n_1 & n_2 & n_3 \\ n_4 & x & n_5 \\ n_6 & n_7 & n_8 \end{pmatrix} \tag{8}$$

$$\begin{cases} G_x = \sqrt{S_x^2 + S_y^2} \\ S_x = (n_3 + 2n_5 + n_8) - (n_1 + 2n_4 + n_6) \\ S_y = (n_1 + 2n_2 + n_3) - (n_6 + 2n_7 + n_8) \end{cases} \tag{9}$$

After determining the edge point and its gradient, you can calculate the direction of the gradient at that point. As shown in Eq. (5), for a point on the circumference, a straight line passing the gradient direction must pass the center of the circle. Calculate

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the number of each point through the gradient straight line, Every time a gradient passes through a coordinate point, the corresponding accumulator of the coordinate increases by one. After setting a threshold, coordinate points larger than the threshold are saved as the coordinates of the center point.

$$\theta = \arctan(\frac{G_y}{G_x}) \tag{9}$$

After determining the coordinates of the center, the distance from the center to each edge point is calculated. The distance from the center to the edge point on the corresponding circle is equal to the radius r_0. Let ***F(d)*** be the number of edge points at a distance d from the center of the circle. ***F(d)*** satisfies Eq. (6) because the point at which the distance from the center of the circle is equal to its corresponding radius must be the largest. By constraining the radius, the number of circles in the image, the region of interest can be identified more accurately and quickly. By constraining the range of radius and the number of circles in the image, the region of interest can be identified more accurately and quickly.

$$\arg\max(F(d)) = r_0 \tag{10}$$

For the acquired hyperspectral image, a band image that is more clearly and clearly detected by the apple can be selected. In this experiment, an image with a wavelength of 1515.83 nm (the 197th band) is selected as a standard image of the region of interest, and then the band Under the image is converted to a grayscale image. For the acquired hyperspectral image, a band image that is more clearly and easily detected by the apple can be selected. In this experiment, an image with a wavelength of 1515.83 nm (the 197th band) is selected as a standard image of the region of interest, and then the image of this band is transformed into a grayscale image. After the grayscale image is obtained, the ROI area is automatically selected according to the Hough circle detection process. In this experiment, the radius of the constrained target circle ranges from 30 to 80 pixels, and the number of target circles is six. ROI regional selection results shown in Fig. 3. Each image containing an apple, after the selection of the region of interest will get two sets of parameters, namely the coordinates of the center of the circle and the length of the radius. The blue circle in the image is the detected area of interest, which is in good agreement with the apple target area.

Considering that apple is not a standard circle, the circle detected by the region of interest cannot completely correspond to the target apple, so when calculating the spectral information, the radius is reduced by half and then the average spectrum is calculated. The circle to be used for calculation is shown in Fig. 4a, and the spectrum curve extracted in Fig. 4a is shown in Fig. 4b, where the abscissa is the wavelength and

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the ordinate is the reflectivity. Use different colors to draw the apple spectral information in different positions. The average spectral information for the two datasets for the 12-h and 96-h sampling is shown in Fig. 5. From the graph, the reflectance of samples taken after 96 h in different wavelengths is lower than the samples sampled after 12 h.

After obtaining the average spectrum of the region of interest, the hyperspectral data is likely to receive external interference during the acquisition, including the noise and environmental factors of the acquisition device itself. This will cause some unrelated interference information to be collected. In order to eliminate the interference noise information while ensuring that the spectral information is not lost as much as possible, data preprocessing is required for obtaining the average spectrum. This experiment uses Savitzky-Golay smoothing to smooth the data. Savitzky-Golay smoothing is a common algorithm for smoothing hyperspectral data.

The Savitzky-Golay smoothing method uses a polynomial to perform polynomial least-squares fitting on the data. In this experiment, a quadratic polynomial is selected for fitting, and the size of the moving window is selected as 15. The smoothing effect of Savitzky-Golay is shown in Fig. 6. Compared with the smoothing result, the smoothing effect is apparent near the bands of 800–900 nm and 1200–1300 nm. Different samples of information are drawn with different colored curves.

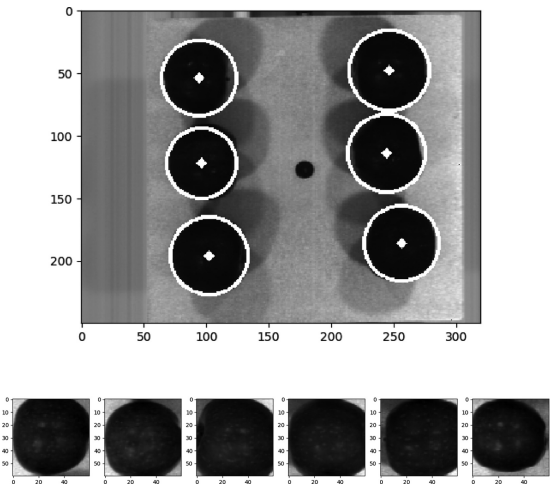
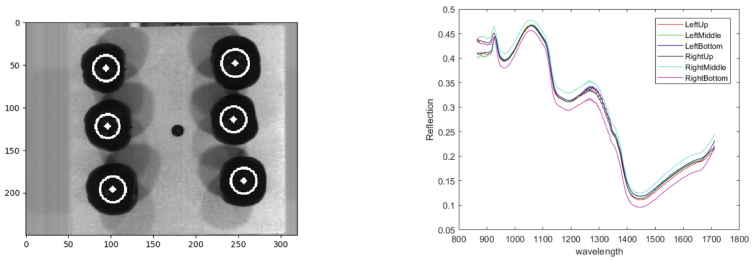


Fig. 3. Schematic diagram of ROI selection (Color figure online)

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a. ROI for calculation

b. calculation results

Fig. 4. Average spectrum acquisition

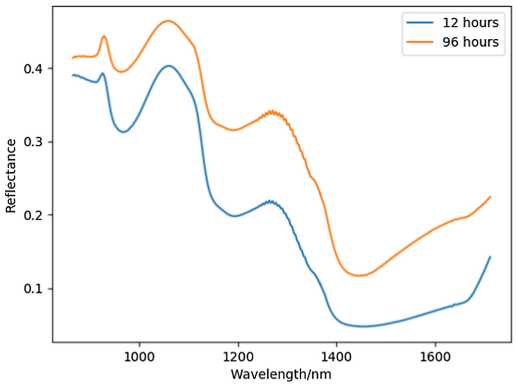


Fig. 5. Comparison of average spectra between data sets

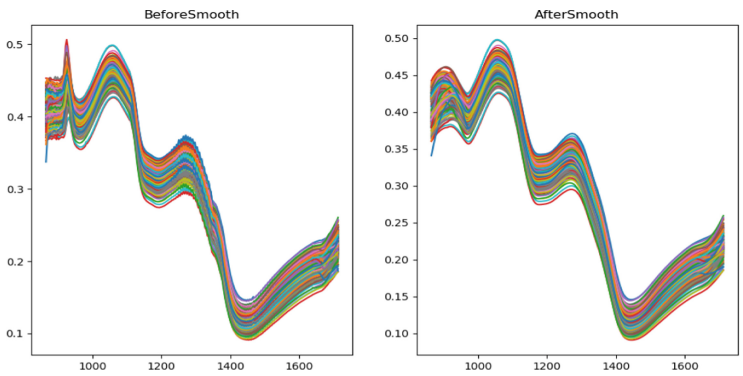


Fig. 6. Savitzky-Golay smoothing diagram

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4 Experimental Results

4.1 Pesticide Residues Detection

4.1.1 Discriminant Modeling on Whole Band

After acquiring the spectral data of the valid region, the experimental data was modeled using Python. We used the above five methods for comparative experiments. When the accuracy rate was calculated, the experiment was conducted using a 5-fold cross validation method. Table 2 shows the results of the experiments with or without pesticide residues.

Table 2. The classification results of pesticide residue in different methods

Method name	Five-fold cross validation accuracy/%	
	96 h	12 h
LDA	98.5	83.5
SVM	99.0	96.5
Fine KNN	94.43	61.1
Decision tree	85.4	64.1
SDE	100.0	95.4

4.1.2 Discriminant Modeling on Feature Band

When using all the bands for modeling, there are some disadvantages such as large data volume and complex modeling. This paper adopts SDE method and SPA method mentioned previously to select the characteristic bands.

When using SPA, set the number of selected bands from 5 to 20. For the discrimination of whether or not there is a pesticide residue, 14 characteristic wavelengths such as a wavelength of 1568.94 nm are selected. Selected wavelengths are shown in Fig. 7 (left). When using the SDE method, we can control the number of selected features. For each problem of pesticide residues detection, we have selected two sets of features with 8 and 18 feature spectra respectively. Figure 8 shows the characteristic bands selected by SDE for pesticide residues.

We also compared the characteristics selected by the SPA and SDE methods. Figure 9 compares the characteristics of the two methods on the issue of whether or not there are pesticide residues. As can be seen from the figure, the two methods generally have the same feature selection and can represent the key information of the spectral curve.

According to the discriminant results on the full band, this experiment selected three methods of LDA, SVM and KNN to detect pesticide residues on the characteristic bands.

The classification results of the three discriminant modeling methods in the full band and different characteristic bands are shown in Table 3. As can be seen from Table 3, the LDA and SVM classifiers use the 18 characteristic bands selected by the SDE to determine whether or not there are pesticide residues give the best performance.

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Table 3. Classification results for pesticide residues based on characteristic wavelength

Method name	Five-fold cross validation accuracy/%	
	96 h	12 h
Full Band + LDA	98.5	83.5
SDE 8 features + LDA	98.98	81.66
SDE 18 features + LDA	100	90.46
SPA 14 features + LDA	100	87.0
Full Band + SVM	99.0	96.5
SDE 8 features + SVM	98.5	64.5
SDE 18 features + SVM	100	95.0
SPA 14 features + SVM	99.5	90.45
Full Band + KNN	94.43	61.1
SDE 8 features + KNN	94.93	62.3
SDE 18 features + KNN	96.43	63.33
SPA 14 features + KNN	93.44	68.72

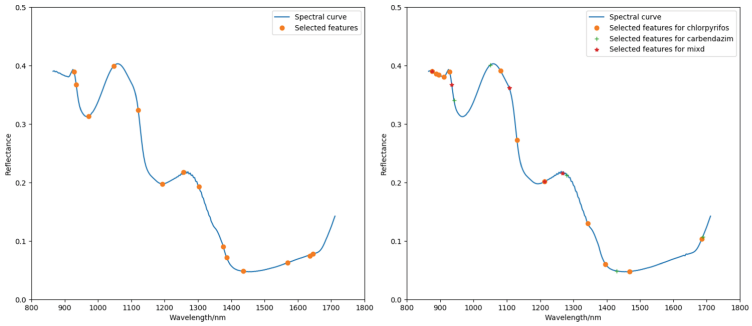


Fig. 7. The feature band selected by the SPA algorithm for pesticide residues (left) and different pesticide residue concentrations (right)

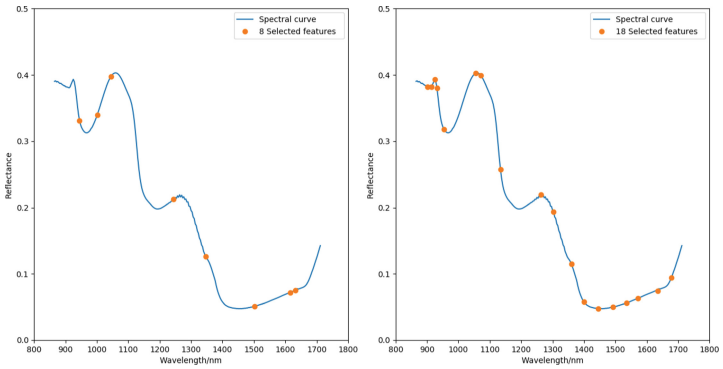


Fig. 8. The feature band selected by the SDE algorithm for pesticide residues (left: 8 features, right: 18 features)

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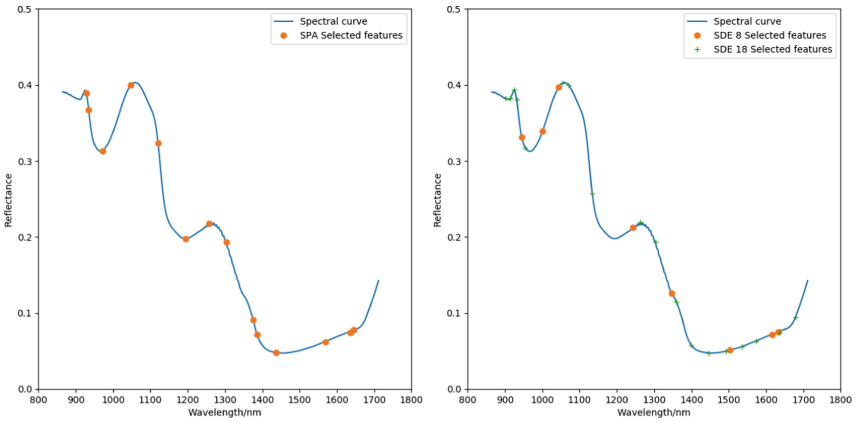


Fig. 9. SPA and SDE feature comparisons on whether or not there are pesticide residues

4.2 Pesticide Residue Concentration Prediction

4.2.1 Discriminant Modeling on Whole Band

After the identification of pesticides on the apple surface, we used five discriminating methods to predict the pesticide residue concentration. The concentrations of the carbendazim group, chlorpyrifos group, and the mixed groups of the two pesticides were discriminated and predicted. The concentration gradient of each pesticide was divided into three gradients: low, medium, and high. Similar to the detection of pesticide residues. When the accuracy rate was calculated, the experiment was also conducted using a 5-fold cross validation method. Table 4 shows the results of the experiments on pesticide residue concentration prediction based on whole band.

Considering the combination of two data sets, Subspace Discriminant has the best performance in the detection of pesticide residue and pesticide concentration. Some test results can reach 95% or more, and the accuracy of some detection items can reach 100%.

Table 4. Classification results of pesticide concentrations in different methods

Method name	Five-fold cross validation accuracy/%					
	96 h			12 h		
	A	B	C	A	B	C
LDA	84.63	69.32	72.35	84.56	82.51	75.47
SVM	71.24	74.1	57.44	76.99	82.66	64.60
Fine KNN	60.90	61.6	51.30	54.30	66.30	38.73
Decision tree	57.42	55.82	46.76	54.48	55.52	37.21
SDE	95.27	84.0	88.90	98.18	96.18	89.09

(A represents Chlorpyrifos, B represents Carbendazim, C represents Mixd)

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4.2.2 Discriminant Modeling on Feature Band

When discriminating the concentration of pesticide residues, SPA selected eight characteristic spectral bands for chlorpyrifos pesticides, five characteristic bands for carbendazim, and eight characteristic bands for the mixed group. SPA selected characteristic bands for different discriminative purposes are shown in Fig. 7 (right).

Figure 10 shows the characteristic bands selected by SDE for different pesticide residue concentrations. The results of pesticide concentration prediction based on characteristic bands are shown in Table 5. In general, the prediction of pesticide concentration based on characteristic bands can achieve relatively high accuracy with a small number of bands. However, the full-band SDE method performs best in pesticide concentration prediction.

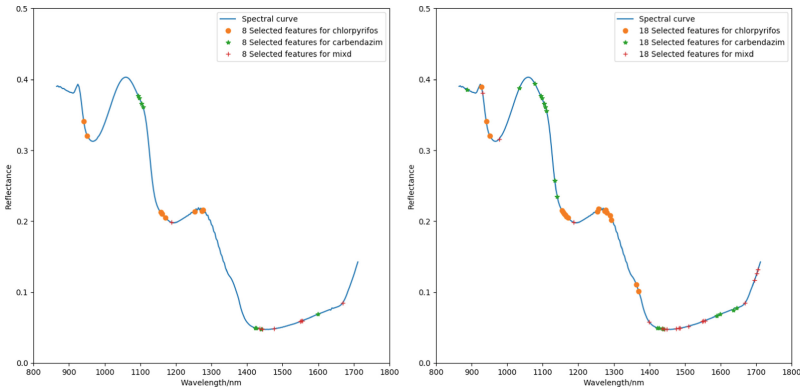


Fig. 10. The feature band selected by the SDE algorithm for different pesticide residue concentrations (left: 8 features, right: 18 features)

Table 5. Classification results for pesticide concentrations based on characteristic wavelength

Method name	Five-fold cross validation accuracy/%					
	96 h			12 h		
	A	B	C	A	B	C
SDE 8 features + LDA	85.55	62.19	65.55	75.50	73.33	73.33
SDE 18 features +LDA	83.33	72.65	65.0	87.47	74.44	77.77
SPA +LDA	81.11	66.34	60.08	79.13	86.66	78.88
SDE 8 features +SVM	63.33	61.11	52.22	55.0	60.0	45.55
SDE 18 features +SVM	71.11	69.06	61.11	56.66	61.66	53.33
SPA +SVM	69.07	68.3	53.26	48.93	63.40	43.42
SDE 8 features +KNN	52.22	58.20	44.31	73.08	66.11	38.33
SDE 18 features +KNN	67.77	60.42	51.66	69.19	70.55	38.88
SPA +KNN	65.21	55.72	56.22	59.19	61.32	39.19

(A represents Chlorpyrifos, B represents Carbendazim, C represents Mixd)

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5 Conclusion

This paper aims at the non-destructive testing of apple pesticide residues, using hyperspectral images to find a suitable modeling method for pesticide residue detection. The subspace discriminant ensemble method has the best effect on the whole band, and the discrimination accuracy of the target apple with respect to pesticides can reach more than 95%. There is also a relatively high accuracy in the identification of pesticide concentration. In feature selection, the SDE algorithm is used to select the feature bands and the traditional SPA algorithm is compared. The experimental results show that in the discriminant modeling of pesticide residues, both methods perform well, in which the use of LDA + SDE18 features, LDA + SPA and SVM + SDE18 features on the data after 96 h of sampling, use 5-fold cross validation can achieve 100% accuracy.

From the comparison of sampling time, apples sampled 96 h after spraying were found to be higher in discrimination accuracy than apples sampled 12 h later. Compared with the prediction of pesticide residue concentration, the accuracy of prediction of pesticide residue concentration of chlorpyrifos is higher than that of carbendazim pesticide. Using the features selected by LDA + SDE to predict chlorpyrifos can reach 85.55%. The predictive effect of the mixture of pesticides has declined. Taken together, the discriminative modeling using the subspace discriminant ensemble method on the whole band can achieve the best results both in the presence of pesticides and in the prediction of pesticide residue concentrations. However, there is still room for improvement in the accuracy of pesticide concentration prediction. This work provides a new idea for the rapid non-destructive testing of apple pesticide residues, and demonstrates that it is feasible to use hyperspectral images to detect pesticide residues.

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